Kubernetes and Istio

09 - First Istio architecture

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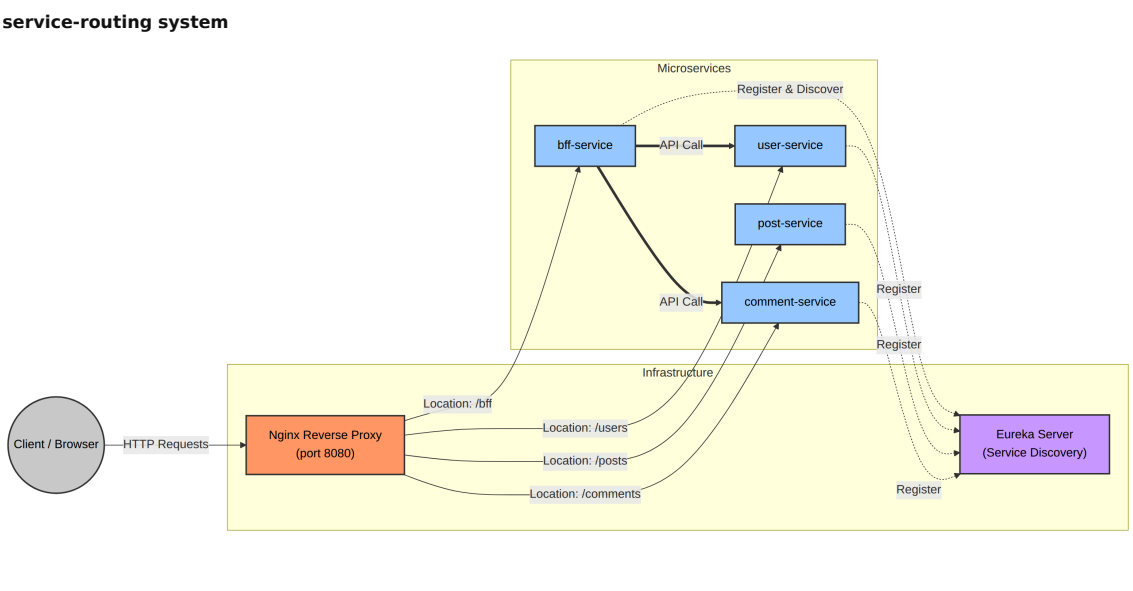
In this lesson, you will learn how to use Istio instead of Eureka and Nginx in the `service-routing` example. Indeed, Kubernetes CoreDNS can seamlessly replace Eureka for service discovery, while the Istio Gateway can replace the Nginx reverse proxy. In addition, Istio provides secure communication between pods (**mTLS**) and powerful observability out of the box.

service-routing system


Microservices
Register & Discover
bff-service
API Call
user-service
post-service
API Call
comment-service
Register
Register
Infrastructure
Location: /bff
Client / Browser
HTTP Requests
Nginx Reverse Proxy (port 8080)
Location: /users
Location: /posts
Location: /comments
Eureka Server (Service Discovery)
Register

Migrating to Kubernetes & Istio

To adapt the existing system to fit an Istio-enabled Kubernetes cluster, we need to perform the following actions:

1. **Remove Nginx Reverse Proxy**: Remove the `nginx` module from your parent build files.
2. **Remove Eureka Service Discovery Server**: Delete the `eureka-service` module completely.
3. **Remove Eureka Dependencies from the microservices**
    1. Remove the `Eureka` and `Spring Cloud` dependencies from `pom.xml`.
    2. Remove Eureka-related configurations from `application.yml`.
    3. Remove any `@LoadBalanced` annotations in the Java codebase (**Kubernetes and Istio will now handle load balancing natively**)
    4. Modify your routing URIs to match the new environment. Thanks to CoreDNS, each microservice can be directly reached at `http://<microservice-name>:<port>`.
4. **Compile the Applications**: Build your Java services using `mvn clean package -Dmaven.test.skip=true`.
5. **Containerize**: Build the Docker images and push them to your container registry.

Define the infrastructure

Now that the container images are ready, we will define the Kubernetes YAML manifests. Each microservice is deployed using a `Deployment` and exposed internally via a `Service`. Here is an example for the `bff-service`:

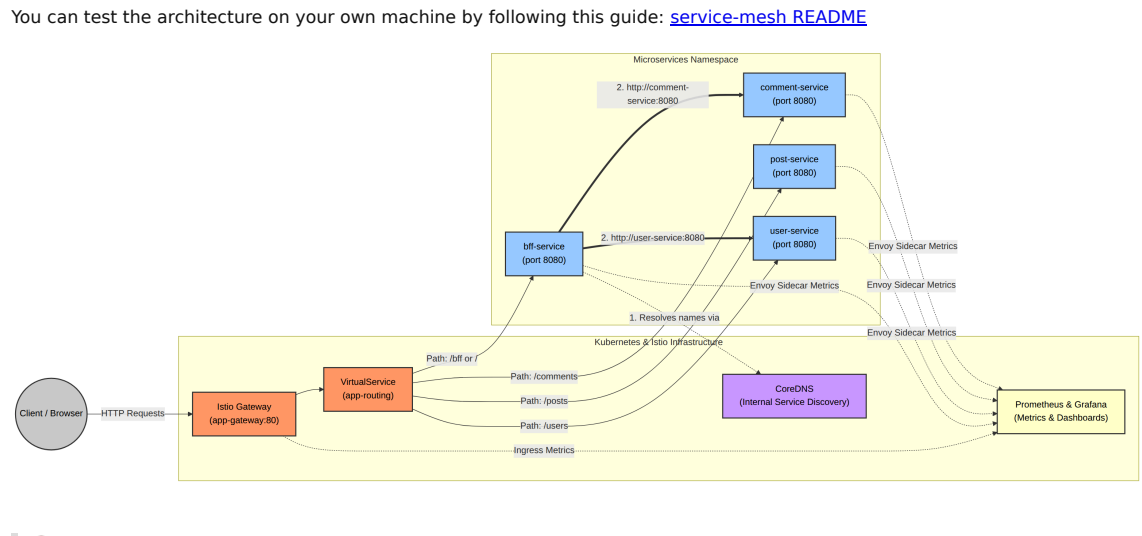
```yaml
apiVersion: v1
kind: Service
metadata:
  name: bff-service
  labels:
    app: bff-service
spec:
  ports:
    - port: 8080
      targetPort: 8080
      name: http
  selector:
    app: bff-service
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: bff-service
spec:
  replicas: 1
  selector:
    matchLabels:
      app: bff-service
  template:
    metadata:
      labels:
        app: bff-service
    spec:
      containers:
        - name: bff-service
          image: mattev02/bff-service:latest
          ports:
            - containerPort: 8080
          env:
            - name: SERVER_PORT
              value: "8080"
          readinessProbe:
            httpGet:
              path: /actuator/health
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 10
          livenessProbe:
            httpGet:
              path: /actuator/health
              port: 8080
            initialDelaySeconds: 20
            periodSeconds: 10
```

To route external traffic into the cluster, we use Istio Gateway API. A `Gateway` handles the entry point (replacing standard Ingress objects), and an `VirtualService` directs the traffic to the corresponding microservices based on the requested path.

```yaml
apiVersion: networking.istio.io/v1
kind: Gateway
metadata:
   name: app-gateway
spec:
   selector:
      istio: ingressgateway
   servers:
      - port:
           number: 80
           name: http
           protocol: HTTP
        hosts:
           - "*"
---
apiVersion: networking.istio.io/v1
kind: VirtualService
metadata:
   name: app-routing
spec:
   hosts:
      - "*"
   gateways:
      - app-gateway
   http:
      # 1. The BFF Route
      - match:
           - uri:
                prefix: /bff
        route:
           - destination:
                host: bff-service
                port:
                   number: 8080

      # 2. The Users Route
      - match:
           - uri:
                prefix: /users
        route:
           - destination:
                host: user-service
                port:
                   number: 8080

      # 3. The Posts Route
      - match:
           - uri:
                prefix: /posts
        route:
           - destination:
                host: post-service
                port:
                   number: 8080

      # 4. The Comments Route
      - match:
           - uri:
                prefix: /comments
        route:
           - destination:
                host: comment-service
                port:
                   number: 8080

      # 5. Default catch-all route fallback
      - route:
           - destination:
                host: bff-service
                port:
                   number: 8080
```

The new architecture

The final system architecture is represented below.
You can test the architecture on your own machine by following this guide: service-mesh README


Microservices Namespace
2. http://comment-service:8080
comment-service (port 8080)
post-service (port 8080)
user-service (port 8080)
bff-service (port 8080)
2. http://user-service:8080
Envoy Sidecar Metrics
Envoy Sidecar Metrics
Envoy Sidecar Metrics
Envoy Sidecar Metrics
1. Resolves names via
Kubernetes & Istio Infrastructure
Path: /bff or /
Path: /comments
Path: /posts
Path: /users
Client / Browser
HTTP Requests
Istio Gateway (app-gateway:80)
VirtualService (app-routing)
CoreDNS (Internal Service Discovery)
Prometheus & Grafana (Metrics & Dashboards)
Ingress Metrics

> ❓ ***What are the advantages of this system compared to the initial one?***
>
> - ***Reduced codebase****: Removes boilerplate code and eliminates the need for maintaining a standalone Eureka server.*
> - ***Language Agnosticism****: While Eureka ties you heavily to the Java ecosystem, Istio works natively with applications written in any language.*
> - ***Offloading Resilience to Infrastructure****: In a traditional Spring Boot app, you must configure libraries like Resilience4j and write fallback logic in Java. Istio manages retries, timeouts, and circuit breaking at the platform level.*
> - ***Simplified Observability****: Metrics, tracing, and logging are captured automatically by the Envoy sidecars.*
> - ***Encrypted Communications****: Istio transparently enforces mTLS between pods, securing internal traffic.*
> - ***Zero-Trust Security****: Istio allows you to define strict* `AuthorizationPolicies`*. For example, you can write a rule stating: "Only bff-service is allowed to call user-service. Reject traffic from any other pod."*